Datasheet

Black Oak Engineering

precision electronic instruments

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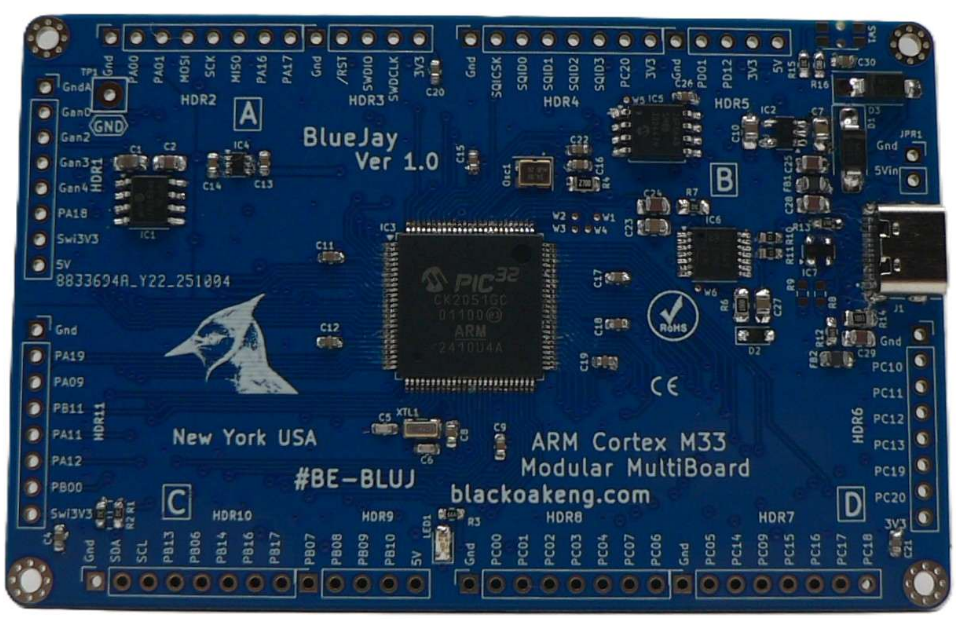
BlueJay ARM Cortex M33

Modular MultiBoard

Part number `BE-BLUJ`
Version `1.0`



Description. The Black Oak Engineering (BOE) BlueJay ARM Cortex M33 Modular MultiBoard (#BE-BLUJ) goes beyond a typical eval or demo board. It is a Single Board Computer (SBC) designed to take on many jobs. It also makes a good development tool. It is compact, but still large enough to be useful. It is based on the popular and widely available ARM Cortex M33 32 bit processor. It has a wide operating temperature range of -40 to +85 °C (-40 to +185 °F). It lends itself well to the development of systems for data acquisition, user interface, AI/ML/Edge, IOT, automation, and OEM integration.

BlueJay is modular. There are four quadrants for placing Wing boards. Each Wing quadrant provides power options and general purpose IO. Each may be physically constrained via M2 mounting holes. The design is completely open, so you may easily design your own Wing board. Wing boards exist for analog data acquisition, battery power, Bluetooth, CAN, Ethernet, RS-232, and WiFi.

A Board Support Package (BSP) and a complete set of various tested drivers are available at our GitHub. BOE releases source code and other collateral under an attached MIT license. BlueJay firmware may be configured under the popular Harmony framework; it also may be configured to support FreeRTOS.

BlueJay has these built-in components:

- SWD interface
- JTAG breakout
- Analog section with 3.00 V Ref.
- Stable primary oscillator.
- 32768 Hz secondary oscillator for RTC.
- Multiple fast SPI & IIC.
- 8Mx8 Flash memory on SQI.
- SQI bus exposed.
- Audio Codec protocols exposed.
- PWM & Position Decoder exposed.
- All IO exposed on 100 mil headers.
- Power from USB or external.
- Multiple power options exposed.
- USB 2.0 Device.
- EMI protections.
- Diagnostic LED.
- Reset switch.
- Made in the USA.

BlueJay may be operated from a standard lithium polymer battery, which BlueJay charges from its power source (USB-C or a header connection).

ARM Cortex M33 specifications (partial)

- 2 MB Flash, 512 KB SRAM,
- Clock to 120 MHz.
- Extensive Hardware Security Module.
- ARM TrustZone.
- Easily programmable in several development ecosystems.

- 4 kB caches.
- Floating Point Unit with DSP instructions.
- 12 bit ADC.
- Peripheral Touch Controller.
- Ethernet MAC, 10/100 Mbps in MII and RMII with dedicated DMA.
- Low power modes.
- CAN & CAN-FD.
- SQI.

BlueJay PCBA specifications

- 90 × 58 mm (3.6 × 2.3 inches).
- 4× 2-56 (or M2) mounting holes.
- Mass = 21.6 g (0.76 oz).

BOE is continuously improving. We also strive to keep one step ahead of procurement shortfalls. We will deliver to you the latest hardware version possible. In some cases specifications will change.

Package includes one 3’ (1 m) USB-C cable.

Environmental

- Temperature. -40 to +85 °C (-40 to +185 °F), excluding battery option.
- Humidity / water exposure. The PCBA does not include a protective enclosure. Nor is it conformally coated. Condensing humidity and water exposure must be completely avoided.

Approvals & Compliance

- RoHS.
- REACH.
- California Prop 65.

Value Added Design. Want to use the BlueJay in a new project or OEM application, but need a little assistance? Not a problem. BOE contracts regularly with end users for value added design.

Warranty Policy. Any instrument ordered from BOE may be returned for full refund, less shipping costs, within 30 days of delivery, provided that the instrument has not, in the opinion of BOE been damaged or misused. An RMA number is required in all cases. See our *Standard Terms & Conditions – Instruments* for more details.

BOE reserves the right to make changes to these specifications as it deems necessary. All technical information contained herein is as accurate as possible; however BOE shall not be held responsible for any errors or for product use, nor for any infringements upon the rights of others which may result from its use. BOE products are not to be used in life support or safety critical applications.

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